Problem K
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Time limit: 10 seconds

You play a game with multiple fair six-sided dice. Each die's face displays a single symbol. The objective of the game is to roll the dice and create a valid word from the symbols on top of each die. If you cannot form a word, you may reroll the dice for another attempt.

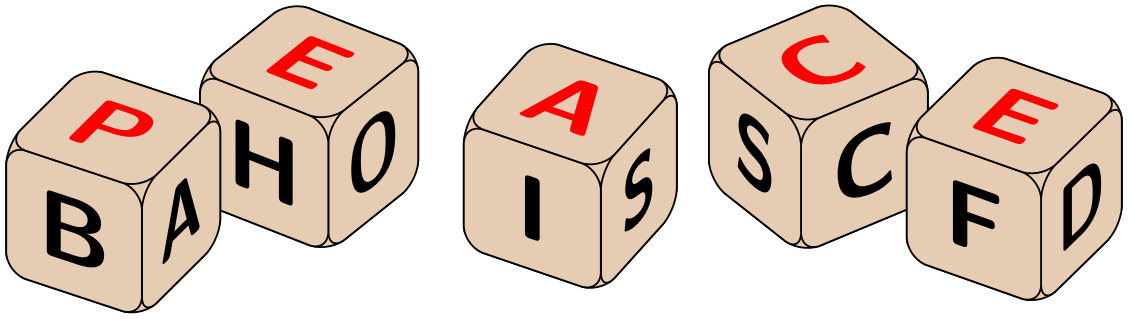


Figure K.1: Five dice making a valid word corresponding to Sample Input 1.

Suppose there are five dice: one of them contains letters `A`, `B`, `C`, `D`, `E`, and `P` (abbreviated as `ABCDEP`), and the other dice contain letters `AEHOXU`, `AISOLR`, `ABCDEF`, and `ABCSCC`. The first roll yields the following letters on the tops of respective dice: `P`, `X`, `R`, `E`, and `S`. As it is impossible to arrange these letters into a valid word, you decide to keep the `P`, `S`, and `E`, and reroll the other dice, in an attempt to make words like `PARSE`, `PAUSE`, `PHASE`, `POISE`, `PROSE`, `PULSE`, or `PURSE`. The two dice yield `E` and `A`, resulting in the following five letters: `P`, `E`, `A`, `E`, and `S`. You still cannot think of a valid word, so you decide to keep four letters and reroll only the last die, which has three sides with letter `C`. By doing so, there is a 50% chance that it will be possible to make a final valid word: `PEACE`, as shown in Figure K.1.

When you roll a die, it lands on any one of its faces with equal probability. What is the expected number of rolls needed to make a valid word, assuming you use an optimal strategy?

Input

The first line of input contains two numbers d and w, where d ($1 \leq d \leq 6$) is the number of dice and w ($1 \leq w \leq 2 \cdot 10^5$) is the number of valid words in the dictionary. The following d lines each have 6 symbols, one for each face of the die. The final w lines contain w distinct valid words in the dictionary. Every word has exactly d symbols.

All symbols in the input are either uppercase letters (`A`–`Z`) or digits (`0`–`9`).

Output

If it is possible to make a valid word, output the expected number of rolls needed to make a valid word when using an optimal strategy. Otherwise, output `impossible`. Your answer should have an absolute or relative error of at most 10^{-6}.

Sample Input 1	Sample Output 1
5 8 ABCDEP AEHOXU AISOLR ABCDEF ABCSCC PARSE PAUSE PHASE POISE PROSE PULSE PURSE PEACE	9.677887141

Sample Input 2	Sample Output 2
2 1 AAAAAA BBBBBB AB	1.0

Sample Input 3	Sample Output 3
3 1 123456 123456 123456 666	10.555444555

Sample Input 4	Sample Output 4
2 1 ABCDEF GHI234 AB	impossible